

HDOC Day 6

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Subject: C Programming

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Question 1: Find Length, Breadth and Diagonal of Rectangle

Aim

Question 1: Find Length, Breadth and Diagonal of Rectangle

Theory

This program calculates the rectangle's dimensions using the given area and perimeter. After finding the length and breadth using mathematical formulas, it calculates the diagonal using the Pythagoras theorem.

Input / Output

Input	Output
Area, Perimeter	Length, Breadth, Diagonal

Formula

Area= $L \times B$

Perimeter= $2(L+B)$

Diagonal= $\sqrt{L^2+B^2}$

Algorithm

1. Start
2. Read area and perimeter
3. Find $s=P/2$
4. Calculate length
5. Calculate breadth
6. Calculate diagonal
7. Display results
8. Stop

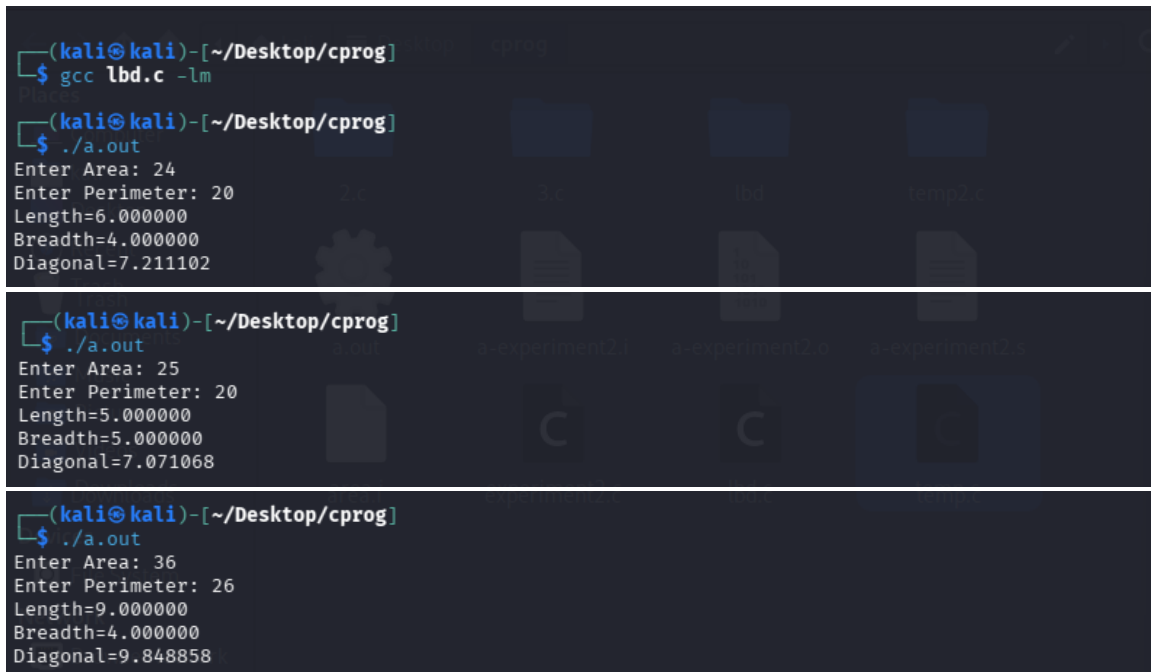
Test Cases

TC	Area	Perimeter	Length	Breadth	Diagonal	Result
1	24	20	6	4	7.21	Pass
2	25	20	5	5	7.07	Pass
3	36	26	9	4	9.85	Pass

C Program

```
#include<stdio.h>
#include<math.h>
int main(){
float A,P,s,l,b,d;
printf("Enter Area: ");
scanf("%f",&A);
printf("Enter the Perimeter: ");
scanf("%f",&P);
s=P/2;
l=(s+sqrt(s*s-4*A))/2;
b=A/l;
d=sqrt(l*l+b*b);
printf("Length=%f\nBreadth=%f\nDiagonal=%f\n",l,b,d);
return 0;
}
```

Screenshots of Compilation and Execution



```
(kali@kali)-[~/Desktop/cprog]
$ gcc lbd.c -lm

(kali@kali)-[~/Desktop/cprog]
$ ./a.out
Enter Area: 24
Enter Perimeter: 20
Length=6.000000
Breadth=4.000000
Diagonal=7.211102

(kali@kali)-[~/Desktop/cprog]
$ ./a.out
Enter Area: 25
Enter Perimeter: 20
Length=5.000000
Breadth=5.000000
Diagonal=7.071068

(kali@kali)-[~/Desktop/cprog]
$ ./a.out
Enter Area: 36
Enter Perimeter: 26
Length=9.000000
Breadth=4.000000
Diagonal=9.848858
```

Conclusion

The program successfully calculates the rectangle dimensions for valid inputs.

Question 2: Remaining Area of Kite after Removing Square

Aim

Question 2: Remaining Area of Kite after Removing Square

Theory

This program computes the area of a kite using its diagonals and subtracts the area of a square cut from it to obtain the remaining area.

Input / Output

Input	Output
Diagonal1, Diagonal2, Square Side	Remaining Area

Formula

Kite Area = $(d1 \times d2) / 2$

Square Area = s^2

Remaining = Kite - Square

Algorithm

9. Start
10. Read diagonals and side
11. Calculate kite area
12. Calculate square area
13. Subtract
14. Display remaining area
15. Stop

Test Cases

TC	D1	D2	Side	Remaining Area	Result
1	20	16	4	144	Pass
2	30	20	5	275	Pass
3	18	12	3	99	Pass

C Program

```
#include<stdio.h>
int main(){
float d1,d2,s,kite,sq,rem;
printf("Enter diagonal 1: ");
scanf("%f",&d1);
printf("Enter the diagonal 2: ");
scanf("%f",&d2);
printf("Enter the side: ");
scanf("%f",&s);
```

```
kite=(d1*d2)/2;  
sq=s*s;  
rem=kite-sq;  
printf("Remaining Kite Area=%f",rem);  
return 0;  
}
```

Screenshots of Compilation and Execution



The first screenshot shows the compilation of the program. The user is in the directory `~/Desktop/cprog` and runs `gcc kite.c -lm`. The second screenshot shows the execution of the program. The user runs `./a.out` and enters the following values: diagonal 1 = 20, diagonal 2 = 16, and side = 4. The output is "Remaining Kite Area= 144.000000". The third screenshot shows another execution of the program. The user runs `./a.out` and enters the following values: diagonal 1 = 30, diagonal 2 = 20, and side = 5. The output is "Remaining Kite Area= 275.000000".

```
(kali@kali)-[~/Desktop/cprog]  
$ gcc kite.c -lm  
  
(kali@kali)-[~/Desktop/cprog]  
$ ./a.out  
Enter the diagonal 1= 20  
Enter the diagonal 2= 16  
Enter the side= 4  
Remaining Kite Area= 144.000000  
  
(kali@kali)-[~/Desktop/cprog]  
$ ./a.out  
Enter the diagonal 1= 30  
Enter the diagonal 2= 20  
Enter the side= 5  
Remaining Kite Area= 275.000000  
  
(kali@kali)-[~/Desktop/cprog]  
$ ./a.out  
Enter the diagonal 1= 18  
Enter the diagonal 2= 12  
Enter the side= 3  
Remaining Kite Area= 99.000000
```

Conclusion

The program correctly calculates the remaining area after removing the square.